SMARTBRIDGE
Let's Bridge the Gap

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	0157CY241012
Project Title	EV Industry Analytics & Intelligence Platform
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To build an interactive EV industry analytics platform (Tableau dashboard and web dashboard) that turns EV market, battery and sustainability data into clear business insights.
Scope	Covers 1,200 EV records (2021-2026) across 12 automakers, 6 vehicle categories and 5 battery chemistries; includes data cleaning, 10+ visualisations, a filterable dashboard, performance testing and GitHub Pages deployment. Real-time vehicle data and ML forecasting are out of scope.
Problem Statement	
Description	EV stakeholders (strategy directors, BI managers and researchers) lack a single, easy-to-use view of sales, market share, battery technology and CO2 impact, and rely on scattered reports.
Impact	Faster, data-driven decisions on product portfolio, OEM benchmarking and battery strategy, and clearer visibility of sustainability benefits such as CO2 displaced.
Proposed Solution	
Approach	Collect and validate the EV dataset, clean and normalise it, compute price-per-km and carbon metrics, build visualisations in Tableau, then integrate them into a responsive web dashboard (HTML5, CSS3,

	Chart.js) hosted on GitHub Pages.
Key Features	Multi-tab dashboard with real-time KPI cards and filters; three persona scenarios (Emily, David, Dr. Sarah); dark/light themes; Tableau Public dashboard; validated data with zero errors; fast queries (1,000 in 7.6 ms).

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard laptop, Intel i5 / Ryzen 5 (quad-core); no GPU required
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	1 GB free SSD space for dataset, workbook and web files
Software		
Frameworks	Python frameworks	HTML5, CSS3, JavaScript, Tableau Desktop / Tableau Public
Libraries	Additional libraries	Chart.js v4.4.2, PowerShell scripts (data generation, export, benchmarking)
Development Environment	IDE, version control	VS Code, Git and GitHub, GitHub Pages
Data		
Data	Source, size, format	Synthetic EV dataset, 1,200 records x 20 columns, CSV / JSON (2021-2026)